

Physics-residual Flow Matching under distribution shift

A study on next-step 2D Navier–Stokes generation

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Abstract

This study measures what a physics-residual penalty does to a Flow Matching model that generates the next state of a 2D Navier–Stokes flow, and how a simple weighted residual compares against the conflict-free formulation used in Physics-Based Flow Matching (PBFM). The comparison is run across 5 seeds, 7 sampling budgets, 4 out-of-distribution shift families with graded severity, and a dense sweep of the residual weight. The results are reported as measured, without selection. Two of them run against the usual expectation. In distribution, a one-line weighted residual (RFM, $\lambda = 0.005$) improves every metric that was tracked, reducing relative L2 by about 37% and the PDE residual by about 44% over the plain baseline, while the parameter-free conflict-free method (PBFM) is statistically indistinguishable from the baseline on all of them. Under the two shifts that actually stress the operator, higher-wavenumber initial conditions and stronger forcing, the residual advantage does not carry over and in several cases reverses: the residual-trained models become less accurate and roll out less stably than the baseline, with PBFM the weakest of the three. The residual is useful only inside a narrow weight band; outside it the model collapses on every axis at once. This is an experimental study that runs the full pipeline, from data generation to evaluation, and it draws its conclusions only from the runs it performed.

1 Motivation and scope

Learned operators for fluid dynamics are by now well established [1], and a generative next-step operator sits naturally on top of that operator view. Physics residuals are a common way to push such a model toward solutions that respect a governing equation [2]. Physics-Based Flow Matching [3] combines a Flow Matching objective with a PDE residual and reports no penalty at inference time, using a conflict-free gradient update [4] in place of a tuned loss weight. The starting question for this study was simple: for a next-step operator on 2D Navier–Stokes, how much does a residual actually change accuracy, physical consistency, distributional fidelity and robustness, and are the extra conflict-free steps worth the added complexity over a plain weighted residual.

The study was set up as a measurement, not a demonstration of a claim. No effect was assumed in advance, no threshold was chosen to be crossed, and no result was framed to favour one method. The value here is in the setup and in reporting what it returns, including the parts that are inconvenient. The three models under test are the plain Flow Matching baseline (FM), the same model with an added weighted physics residual (RFM), and the PBFM recipe (conflict-free update, sample unrolling and time-weighted residual). The data, model, sampler and budget are held fixed across the comparison. Two implementation details of the PBFM branch also differ and are incidental: it runs in float32 and does not clip gradients, which its conflict-free update requires, whereas FM and RFM use mixed precision with the gradient norm clipped to 1.0.

2 Problem setup and method

2.1 The generative task

Flow Matching is used here as a conditional next-step operator. Given a vorticity field converted to a divergence-free velocity u_t and a Reynolds number Re , the model generates the next state $u_{t+\Delta t}$ from Gaussian noise. The conditional optimal-transport path is $x_\tau = (1-\tau)x_0 + \tau x_1$ with target velocity $x_1 - x_0$, and the network $v_\theta(x_\tau, \tau \mid u_t, \text{Re})$ regresses that velocity [5]. Sampling integrates the learned field with an explicit Euler solver whose number of function evaluations (NFE) is counted exactly. Because the target next state is fixed given (u_t, Re) , pointwise error against that target is well defined, which is what makes the accuracy axis meaningful for a generative model.

2.2 The physics residual

The residual is evaluated on the predicted endpoint $\hat{x}_1 = x_\tau + (1-\tau)v_\theta$, never on an interpolated point on the path, since the interpolation has no reason to satisfy the equation. It is the full vorticity-transport residual of the 2D Navier–Stokes equation together with an incompressibility term, weighted linearly in τ . Both terms are divided by a fixed ground-truth scale so that a single weight is comparable across the Reynolds range. This is the dynamical constraint appropriate to a next-step operator and is the analogue of the full-PDE residual case in the PBFM paper rather than the snapshot-only case. The governing equation and the exact form of the residual are written out in Appendix A.

2.3 Three ways to combine the two objectives

The Flow Matching gradient g_{FM} and the physics gradient g_{R} can disagree in direction. The three models differ in how they combine the two gradients (Figure 1). FM ignores the residual. RFM adds it with a scalar weight, $g = g_{\text{FM}} + \lambda g_{\text{R}}$. PBFM uses the conflict-free update of ConFIG [4], which builds a direction that has a positive inner product with both gradients, unlike a projection method such as gradient surgery [6], and therefore descends both objectives at once, with no weight to tune. With $O(a, b) = b - \frac{a \cdot b}{\|a\|^2} a$ the component of b orthogonal to a and $U(g) = g/\|g\|$,

$$g_v = U(U(O(g_{\text{FM}}, g_{\text{R}})) + U(O(g_{\text{R}}, g_{\text{FM}}))), \quad g_{\text{PBFM}} = (g_{\text{FM}} \cdot g_v + g_{\text{R}} \cdot g_v) g_v. \quad (1)$$

PBFM additionally unrolls the sample toward $\tau = 1$ over a few solver steps before evaluating the residual, which reduces the gap introduced by estimating the endpoint from a single step, and weights the residual by τ . The two-gradient ConFIG update above is computed in closed form, which is exact for two objectives, because the reference implementation’s least-squares step failed on the GPU for the full parameter vector.

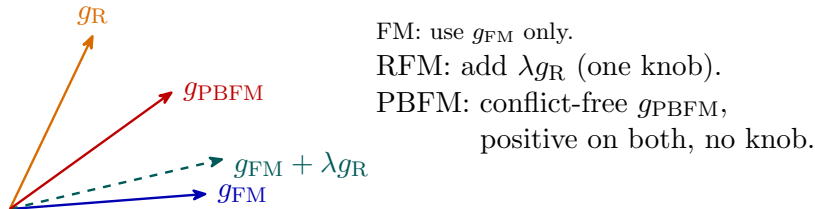


Figure 1: The three training modes differ only in how the Flow Matching and physics gradients are combined. With a small λ the weighted sum stays close to g_{FM} ; the conflict-free direction is constructed to descend both objectives.

2.4 Evaluation

Four families of measurement are recorded for every trained model, each computed per sampling budget and per shift regime.

- Distributional fidelity $\mathcal{D}_{\text{data}}$: kernel MMD [7] (RBF kernel with the median-distance bandwidth, biased estimator) and sliced Wasserstein distance, both on the velocity fields average-pooled to 16×16 .
- Pointwise fidelity $\mathcal{D}_{\text{sol}}$: relative L2 against the true next state.
- Physical fidelity $\mathcal{D}_{\text{phys}}$: root-mean-square transport residual and RMS divergence.
- Rollout stability: the operator is fed its own output as the next condition, and the horizon at which relative L2 first exceeds 0.3 is recorded as t_{fail} (capped at the rollout length of 20 steps).

Out-of-distribution is kept as 4 separate axes rather than one score. A severity $\delta \geq 0$ moves each shift away from training: *param* extrapolates Reynolds number ($\text{Re} = \text{Re}_{\text{max}}(1 + \delta)$), *ic* raises the initial-condition wavenumber, *forcing* scales the forcing amplitude, and *resolution* evaluates on a finer grid. Every out-of-distribution regime is normalised with the training statistics, not its own, so the shift is not hidden by renormalisation. The severity δ and the exact map for each family are defined in Appendix B.

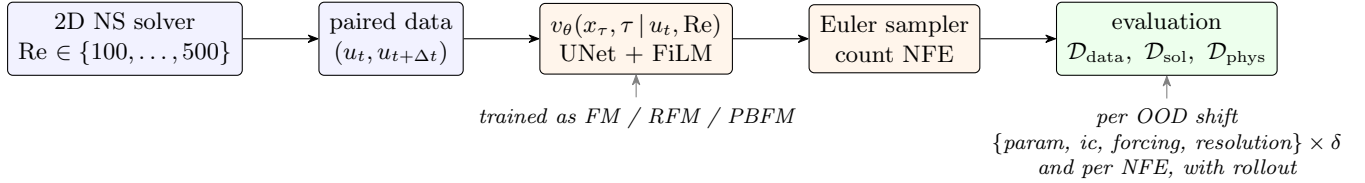


Figure 2: The pipeline. A pseudospectral solver produces trajectories across a training band of Reynolds numbers; paired next-step samples train a conditional velocity field under one of the three modes; an Euler sampler with a counted budget generates states that are scored on three axes across the shift grid and under autoregressive rollout.

3 Experimental protocol

Data are generated by a pseudospectral solver on a 64×64 periodic grid with Kolmogorov forcing, at Reynolds numbers $\{100, 200, 300, 400, 500\}$, with $\text{Re} = 300$ as the in-distribution evaluation point. The velocity network is a conditional UNet of width 48 with feature-wise modulation on the diffusion time and on Reynolds number (Appendix C). Training runs for 100 epochs with AdamW at learning rate 2×10^{-4} , batch size 32, cosine decay, mixed precision and an exponential moving average of the weights. The residual weight for the model comparison is $\lambda = 0.005$, inside the working band established by the sweep in Section 4.4. Sampling uses Euler throughout so that step count and NFE coincide and no comparison is confounded by the solver. The full configuration is listed in Appendix D.

The model comparison covers the three modes over 5 seeds. The residual-weight sweep covers 8 values of λ over 3 seeds. Every number reported below is a mean over seeds, and the intervals are 95% bootstrap confidence intervals over the seed population. This seed count is small, and where it produces wide intervals the study says so rather than smoothing over it.

4 Results and observations

This section is organised one subsection per measurement, and the figures carry the main content. Every figure caption states what the plot shows and what is read off it.

4.1 Training behaviour

Figure 3 shows how training proceeds. Both losses fall within about 20 epochs and are flat by 100, so all three models reach convergence on the same budget. The cosine between the Flow Matching gradient and the physics gradient stays mostly positive through training, in a band from about -0.2 to $+0.7$ with a mean near $+0.2$. The two objectives point in broadly similar directions during training rather than opposing each other, which is the reference point for reading the residual-weight collapse in Section 4.4.

Training dynamics

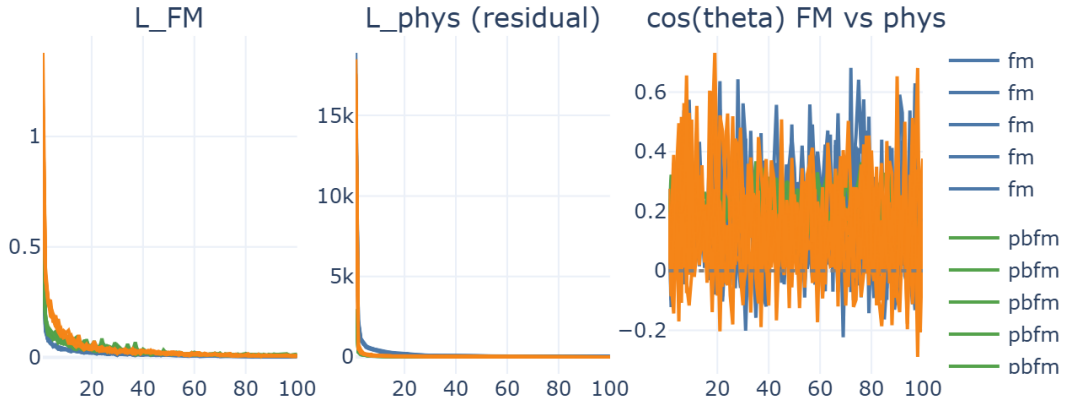


Figure 3: Training dynamics over 100 epochs. Observed: the Flow Matching loss (left) and the physics residual loss (middle) both drop steeply and settle by epoch 20. The cosine between the two gradients (right) fluctuates but stays mostly above zero (roughly -0.2 to $+0.7$), so the FM and physics objectives are not in strong directional conflict on this task.

4.2 In distribution the weighted residual wins and the conflict-free method does not

Table 1 reports the three models at the evaluation budget of 16 Euler steps. The weighted residual improves every axis at once. Relative L2 falls from 0.0148 to 0.0094, a reduction of about 37%; the transport residual falls from 13.2 to 7.5, about 44%; MMD falls by about 63% and RMS divergence by about 41%. The conflict-free model tells the opposite story. On relative L2, residual, MMD, divergence and sliced Wasserstein its confidence interval overlaps the baseline in every case, so within the precision of 5 seeds PBFM is not separable from plain Flow Matching. The added parts of the recipe, the conflict-free update with unrolling and time weighting, gave no measurable in-distribution gain here, while the one-line weighted penalty gave a large one.

The same ordering holds at a matched sampling budget and on the physical axis. Figure 5 repeats the comparison at a fixed 16 steps with confidence intervals: RFM is lowest on both

Table 1: In-distribution metrics at $Re = 300$ and 16 Euler steps. Mean over 5 seeds with 95% bootstrap half-width. Lower is better for all columns. Best value in each column in **bold**.

model	relative L2	residual	MMD ($\times 10^{-3}$)	divergence	SWD ($\times 10^{-3}$)
FM	0.0148 ± 0.0021	13.22 ± 1.96	0.62 ± 0.26	0.125 ± 0.019	3.45 ± 0.52
RFM	0.0094 ± 0.0008	7.47 ± 0.87	0.23 ± 0.06	0.073 ± 0.008	2.36 ± 0.18
PBFM	0.0146 ± 0.0017	12.17 ± 1.58	0.66 ± 0.25	0.118 ± 0.016	3.54 ± 0.55

In-distribution relative L2 at NFE=16

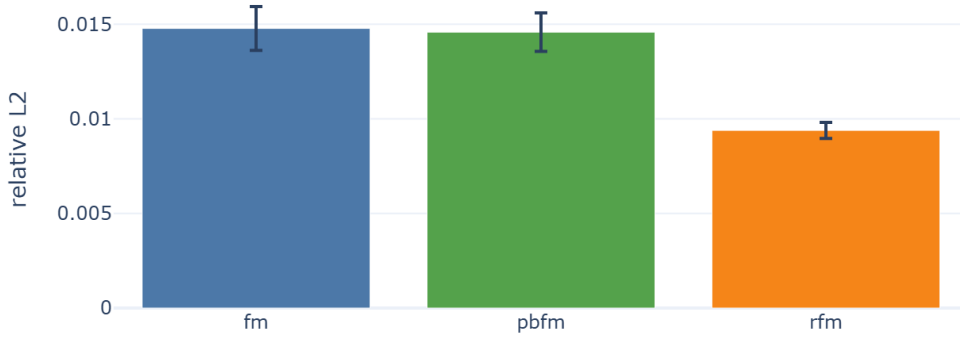


Figure 4: In-distribution quality of the three models at 16 steps. Observed: RFM has the lowest bar on every axis (relative L2 0.0094 against 0.0148 for FM and 0.0146 for PBFM; residual 7.5 against 13.2 and 12.2), while the FM and PBFM bars sit within each other’s confidence intervals on all five metrics. RFM is the only model that moves on the accuracy, physical and distributional axes at the same time.

relative L2 and PDE residual, and the FM and PBFM intervals overlap. Two views of a generated field show where the error sits. Figure 6 plots the pointwise PDE residual of a generated sample and the radial energy spectrum. The residual is grid-scale noise spread across the domain rather than a coherent large-scale error, and the generated spectrum tracks the reference up to wavenumber $k \approx 13$ then flattens near 10^{-6} while the true spectrum keeps decaying, so the model adds high-wavenumber energy that the true dynamics do not contain.

Matched-budget comparison at 16 steps (lower is better)

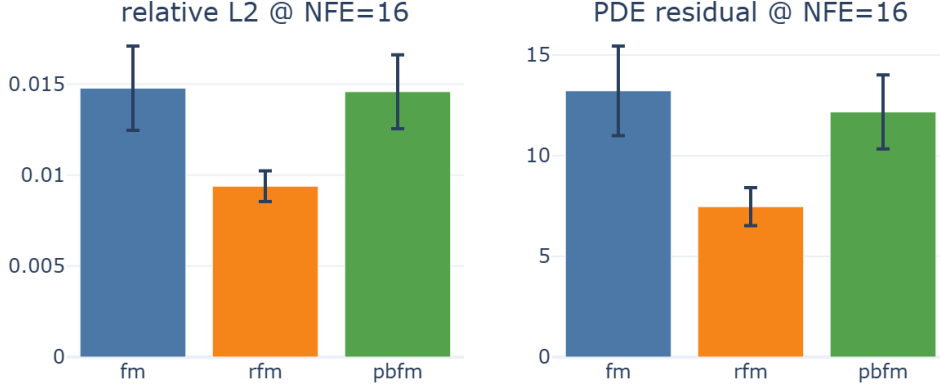
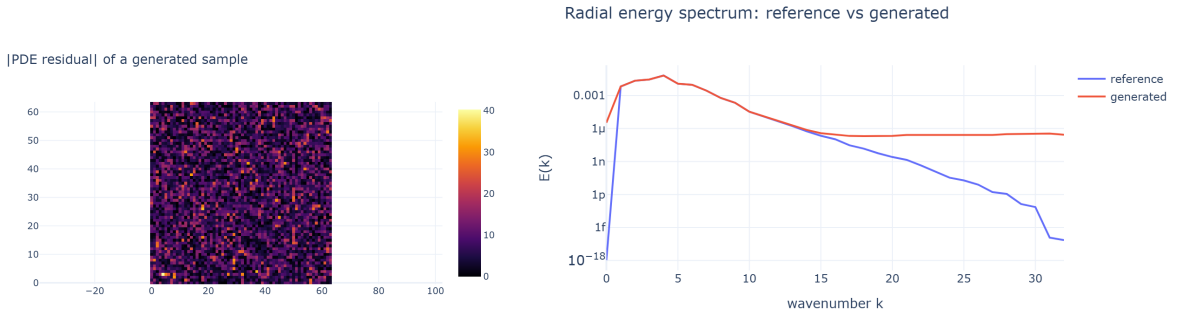


Figure 5: In-distribution comparison at a matched budget of 16 Euler steps. Observed: RFM has the lowest relative L2 (left) and the lowest PDE residual (right), and its error bar clears the other two, while FM and PBFM overlap on both axes. Fixing the budget removes sampling cost as a confound.



(a) Pointwise |PDE residual|.

(b) Radial energy spectrum.

Figure 6: Where the in-distribution error lives. Observed: the residual field (left) is fine, grid-scale noise spread over the whole domain rather than a few coherent hot spots. The spectrum (right) matches the reference up to $k \approx 13$, then the generated curve flattens at about 10^{-6} while the reference decays several more orders of magnitude, so the model adds small-scale energy that the true field does not have.

4.3 Sampling budget: the residual reaches the baseline accuracy with far fewer steps

Every model improves monotonically as the sampling budget grows from 1 to 64 steps (Figure 7, Table 2). The in-distribution ordering holds at every budget: RFM is best throughout and FM and PBFM track each other. The practical consequence is a large efficiency gap. The baseline needs its full 64-step budget to reach a relative L2 of 0.0113; the weighted residual reaches the same accuracy in 4 to 8 steps, about 1/8 to 1/16 of the compute for the same pointwise error. The conflict-free model gives a small edge over the baseline only at the highest budgets, and the intervals overlap even there.

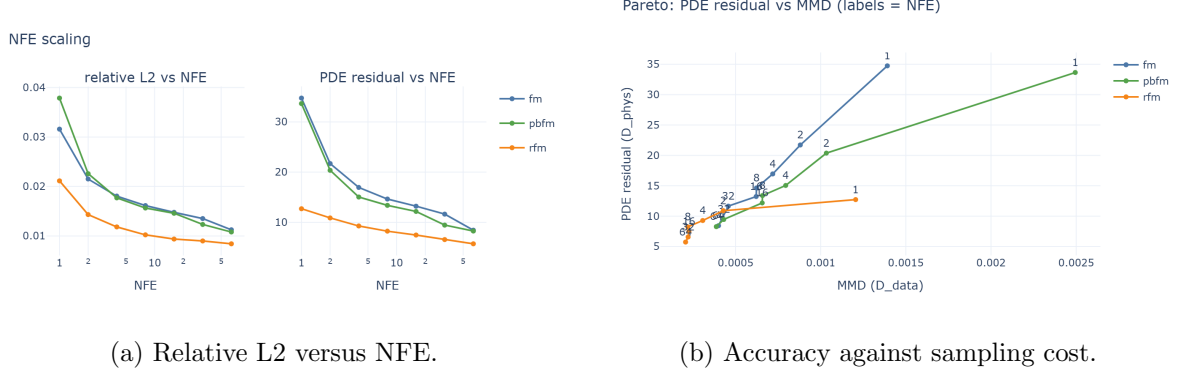


Figure 7: Sampling budget. Observed: relative L2 falls monotonically with steps for all three (left), and the RFM curve lies below the other two at every budget, reaching 0.0118 by 4 steps, which FM only reaches near 64 steps. The accuracy-against-cost view (right) puts RFM on a strictly better frontier, while FM and PBFM overlap across the whole range.

Table 2: Relative L2 against the number of Euler steps (NFE). Mean over 5 seeds.

model	1	2	4	8	16	32	64
FM	0.0316	0.0215	0.0181	0.0161	0.0148	0.0135	0.0113
RFM	0.0211	0.0143	0.0118	0.0102	0.0094	0.0090	0.0084
PBFM	0.0379	0.0226	0.0177	0.0157	0.0146	0.0123	0.0108

4.4 Residual weight has a narrow working band and a hard collapse

The residual is not a free improvement that grows with its weight. Table 3 and Figure 8 sweep λ from 0 to 0.1. At $\lambda = 0$ the model reproduces the Flow Matching baseline, which is a useful internal check. The best accuracy sits at λ between 0.005 and 0.01. From $\lambda = 0.02$ onward every axis degrades together, and by $\lambda = 0.1$ the relative L2 is 0.0503, which is 5.4x the optimum and 3.4x the baseline. The physical residual itself is worst at the largest weight (25.3 against 7.2 at the minimum), so over-weighting the physics term makes the physics worse, because the sample quality it depends on has collapsed. The usable range is roughly a factor of 10 wide, and it has to be found rather than assumed.

Sensitivity to the residual weight

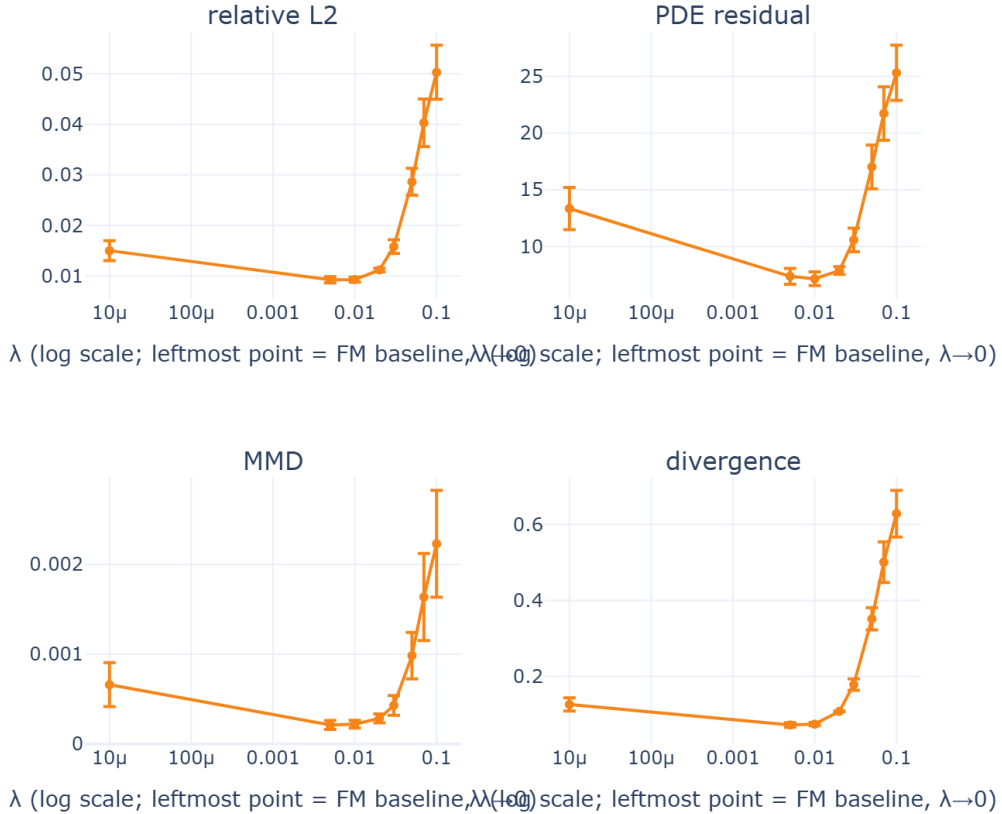


Figure 8: Sensitivity to the residual weight. Observed: all four axes reach their minimum at λ between 0.005 and 0.01 and rise together past 0.02. At $\lambda = 0.1$ relative L2 is 0.0503 (about 5.4x the minimum) and the residual is 25.3 against 7.2 at the minimum, so the physics term is worst exactly where it is weighted most.

Table 3: Residual-weight sweep at 16 Euler steps (RFM, mean over 3 seeds). The minima on every row fall at $\lambda \in [0.005, 0.01]$.

λ	0	0.005	0.01	0.02	0.03	0.05	0.07	0.1
relative L2	0.0150	0.0093	0.0093	0.0112	0.0158	0.0286	0.0403	0.0503
residual	13.34	7.37	7.16	7.88	10.58	17.02	21.73	25.31
MMD ($\times 10^{-3}$)	0.66	0.21	0.22	0.29	0.43	0.98	1.64	2.23
divergence	0.126	0.072	0.074	0.108	0.178	0.352	0.501	0.629

4.5 Out of distribution the residual advantage does not carry over, and often reverses

The out-of-distribution results are where the in-distribution story breaks. They are summarised in Table 4 and Figures 9 and 10.

Reynolds extrapolation does not stress the operator. Out to $\delta = 2$, which is $Re = 1500$ or 3x the training maximum, no model’s one-step error rises above its in-distribution value, and every rollout survives the full horizon. The param axis, as constructed, does not function as a stressor in this range, because a higher Reynolds number at fixed forcing and resolution produces qualitatively similar fields and the model is told the new value through its conditioning.

Higher-wavenumber initial conditions reverse the ordering. The moment the initial-condition wavenumber is raised, error jumps for all three models, and past $\delta = 1$ the residual-trained models are worse than the baseline, not better. At $\delta = 2$ the baseline sits at relative L2 0.052 while RFM is at 0.069 and PBFM at 0.078. The model that was 37% more accurate in distribution is 33% less accurate under this shift, and the conflict-free model is the weakest of the three. The seed spread is also much larger for the residual models here: the PBFM interval at $\delta = 2$ runs from 0.033 to 0.143, a factor of 4, so the residual models are not only worse in the mean but less reliable across seeds.

Stronger forcing breaks accuracy for everyone and rollout stability for the residual models first. One-step error grows steeply with forcing amplitude for all three, and at $\delta = 2$ RFM is marginally ahead of the baseline on one-step error while PBFM is behind. The rollout tells a sharper story: at $\delta = 2$ the baseline holds out to step 15.6 while RFM fails at step 9.6 and PBFM at 11.2. Under this shift the residual buys a little one-step accuracy and pays for it with markedly earlier rollout divergence.

Resolution is coarse and noisy. The resolution axis rounds severity to an integer grid multiplier, so it moves in large jumps and the values plateau once the multiplier saturates. The residuals inflate with grid size as expected, the one-step errors are within noise of each other, and no clean ordering emerges. This axis carries little signal in its current form.

One point about the study design is worth stating plainly. The one-step breaking point, defined as the severity at which relative L2 first passes 0.3, stays pinned at $\delta = 2$ for every model and every category, because the largest one-step error seen anywhere is about 0.18. The threshold was never reached, so that measure does not tell the models apart. The signal that does is the rollout horizon t_{fail} , which separates the models under ic and forcing.

The reversal is not confined to relative L2. Figure 11 plots MMD and transport residual against severity, and both show the same crossing: under ic and forcing the residual models rise above FM at high severity. At ic $\delta = 2$ the MMD is 0.011 for FM against 0.034 for RFM and 0.029 for PBFM. Figure 12 normalises each model by its own in-distribution value, which removes the different starting points and shows relative fragility directly. FM degrades by about $\times 2.7$ under both ic and forcing, while RFM and PBFM degrade by $\times 4.5$ to $\times 5.7$, so measured against where each model starts, the residual models are close to twice as fragile to these two shifts. The param column sits near $\times 1$ for all three, confirming that Reynolds extrapolation

OOD degradation: rel_l2

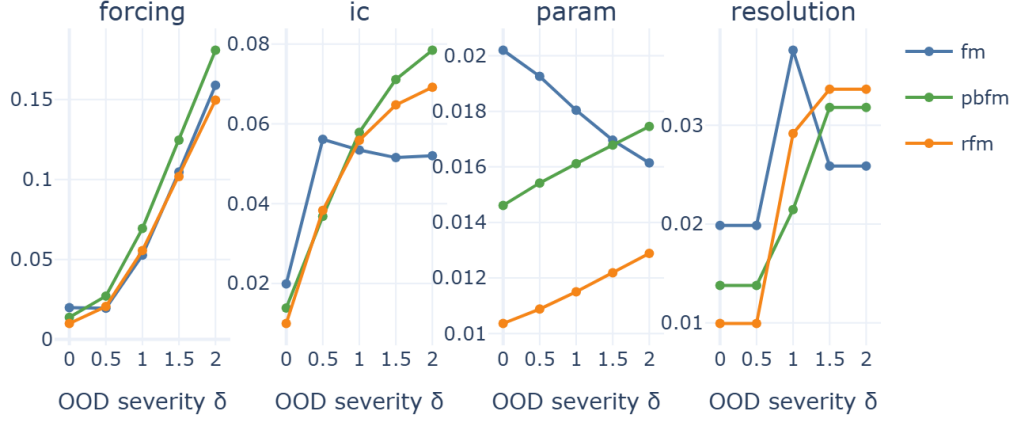


Figure 9: Relative L2 against severity for each shift family. Observed: the param curves stay flat out to $\delta = 2$ (Re 1500). The ic and forcing curves rise, and past $\delta = 1$ the RFM and PBFM curves cross above the FM curve. At forcing $\delta = 2$ the order is RFM 0.150, FM 0.159, PBFM 0.181; at ic $\delta = 2$ it is FM 0.052, RFM 0.069, PBFM 0.078. The model that led in distribution trails under these two shifts.

Relative L2 across OOD category $\times$ severity

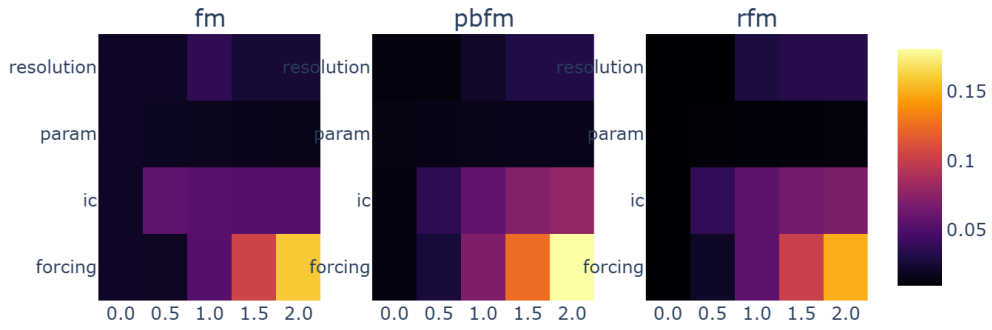


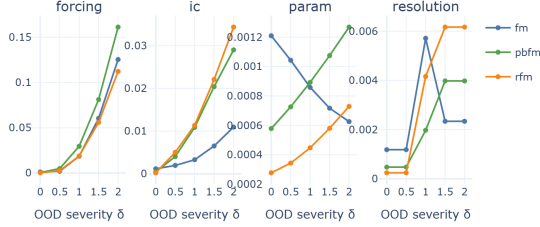
Figure 10: Robustness across the full shift grid, with darker cells marking larger error. Observed: the param and resolution columns stay light across all severities, while the ic and forcing columns darken from $\delta = 1$ onward, and they darken faster for RFM and PBFM than for FM. The map makes the two effective shifts and the residual models' greater sensitivity to them visible at a glance.

Table 4: Out-of-distribution relative L2 (top) and rollout t_{fail} (bottom) by category and severity, mean over 5 seeds. For t_{fail} , higher is better and 20 is the rollout cap. Values at $\delta = 0$ coincide with the in-distribution point.

		severity δ				
metric / category	model	0.0	0.5	1.0	1.5	2.0
<i>relative L2</i>						
ic	FM	0.020	0.056	0.053	0.052	0.052
	RFM	0.010	0.038	0.056	0.065	0.069
	PBFM	0.014	0.037	0.058	0.071	0.078
forcing	FM	0.020	0.020	0.053	0.105	0.159
	RFM	0.010	0.021	0.056	0.102	0.150
	PBFM	0.014	0.027	0.069	0.125	0.181
param	FM	0.020	0.019	0.018	0.017	0.016
	RFM	0.010	0.011	0.012	0.012	0.013
	PBFM	0.015	0.015	0.016	0.017	0.018
<i>rollout t_{fail}</i>						
ic	FM	20.0	18.0	15.8	12.2	10.2
	RFM	20.0	13.4	11.2	10.0	9.2
	PBFM	20.0	13.2	11.2	10.8	10.2
forcing	FM	20.0	20.0	20.0	20.0	15.6
	RFM	20.0	20.0	20.0	17.0	9.6
	PBFM	20.0	20.0	20.0	19.4	11.2

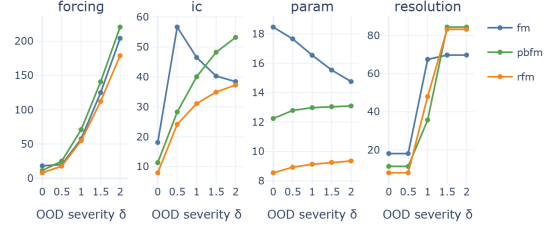
does not move any model.

OOD degradation: mmd



(a) MMD against severity.

OOD degradation: residual



(b) Transport residual against severity.

Figure 11: Out-of-distribution degradation on the distributional and physical axes. Observed: both metrics stay flat under param, rise under ic and forcing, and past $\delta = 1$ the RFM and PBFM curves cross above FM, the same ordering seen in relative L2. The distributional and physical error move together with the pointwise error.

OOD degradation ratio: $\text{relL2}(\delta=1) / \text{relL2}(\delta=0)$

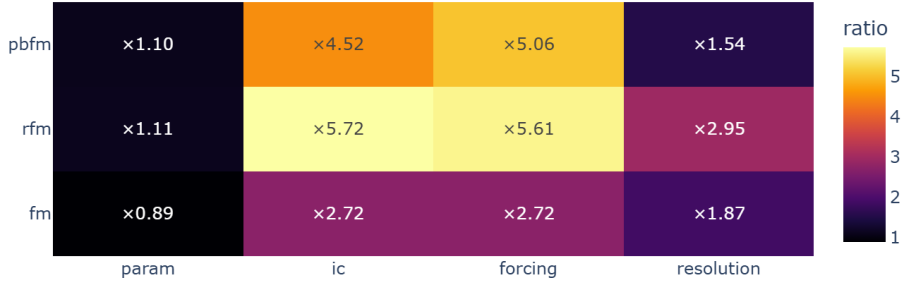


Figure 12: Degradation ratio, each model’s relative L2 at $\delta = 1$ divided by its own value at $\delta = 0$, so higher means it degrades more from its own baseline. Observed: under ic and forcing FM degrades by $\times 2.7$ while RFM reaches $\times 5.7$ and $\times 5.6$ and PBFM $\times 4.5$ and $\times 5.1$. The param column is near $\times 1$ (FM even slightly below 1). This normalised view isolates the reversal from the fact that the residual models start lower in distribution.

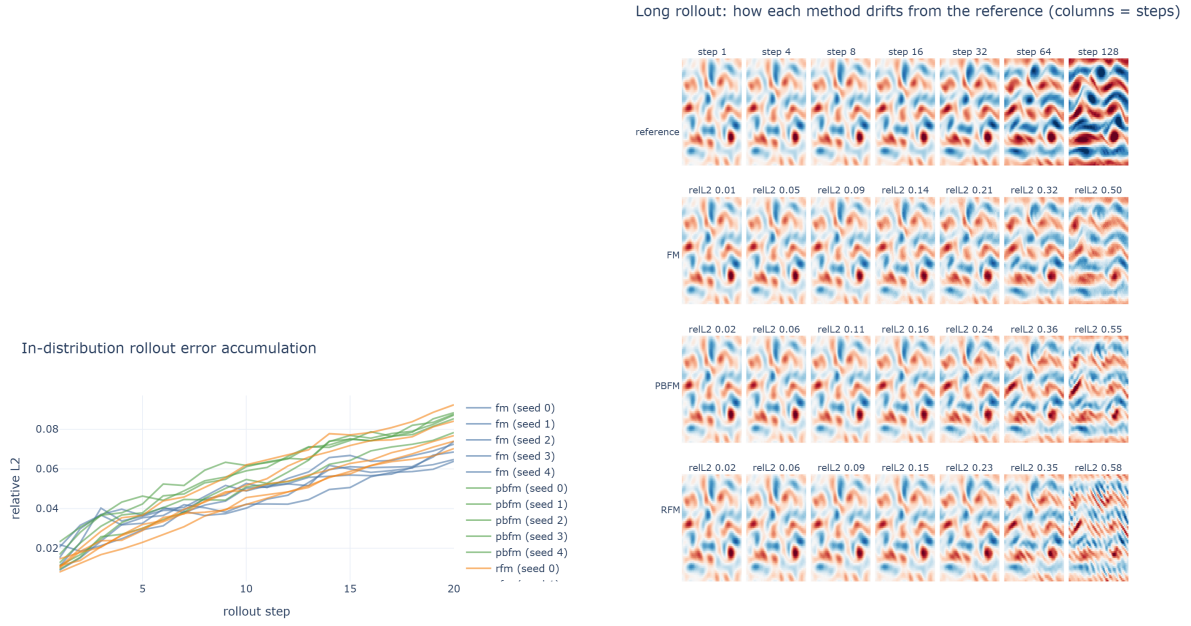
4.6 Rollout

The rollout results reinforce the same point from a different angle. In distribution and under param and resolution shift, all three models survive the full 20-step horizon. Under ic shift the baseline degrades the most gracefully, losing roughly 2 steps of stable horizon per 0.5 of severity, while the residual models drop faster and level off lower. Under forcing shift the gap opens only at the strongest severity, where the residual models fail 6 to 9 steps earlier than the baseline (at forcing $\delta = 2$, RFM fails at step 9.6 and PBFM at 11.2 against the baseline’s 15.6). Whatever the residual does to sharpen a single step, it does not translate into longer stable rollouts once the input distribution moves.

4.7 Qualitative samples

The generated fields are consistent with the metrics rather than an independent story. In distribution the three models produce vorticity fields close to the reference, with the residual model visibly sharper at small scales, which matches its lower spectral and distributional error. Under the ic and forcing shifts the generated fields depart from the reference in the same order the metrics report, and the residual models show more small-scale structure that is present but misplaced, which is consistent with a model that has learned the training physics well and applies it confidently to inputs it was not trained on.

Figure 15 quantifies the same four samples. It reports the per-sample relative L2 and PDE residual for each method on identical conditioning fields and identical source noise, so the differences are the method’s and not sampling luck. RFM has the lowest bars on most of the samples on both axes, consistent with the grids above and with the in-distribution tables.



(a) Relative L2 along the rollout.

(b) Generated fields along a 128-step rollout.

Figure 13: Autoregressive rollout. Observed: in distribution all three curves stay under the failure threshold to the 20-step cap (left). Under forcing at $\delta = 2$ the FM curve holds until step 15.6 while RFM crosses at 9.6 and PBFM at 11.2. The generated fields (right) roll a single in-distribution trajectory out to 128 steps. FM has the lowest relative L2 at every step (0.50 at step 128 against 0.55 for PBFM and 0.58 for RFM), and by step 128 the RFM field breaks into grid-scale striping, with the extra small-scale energy from Figure 6 growing into visible noise, while FM stays smoother.

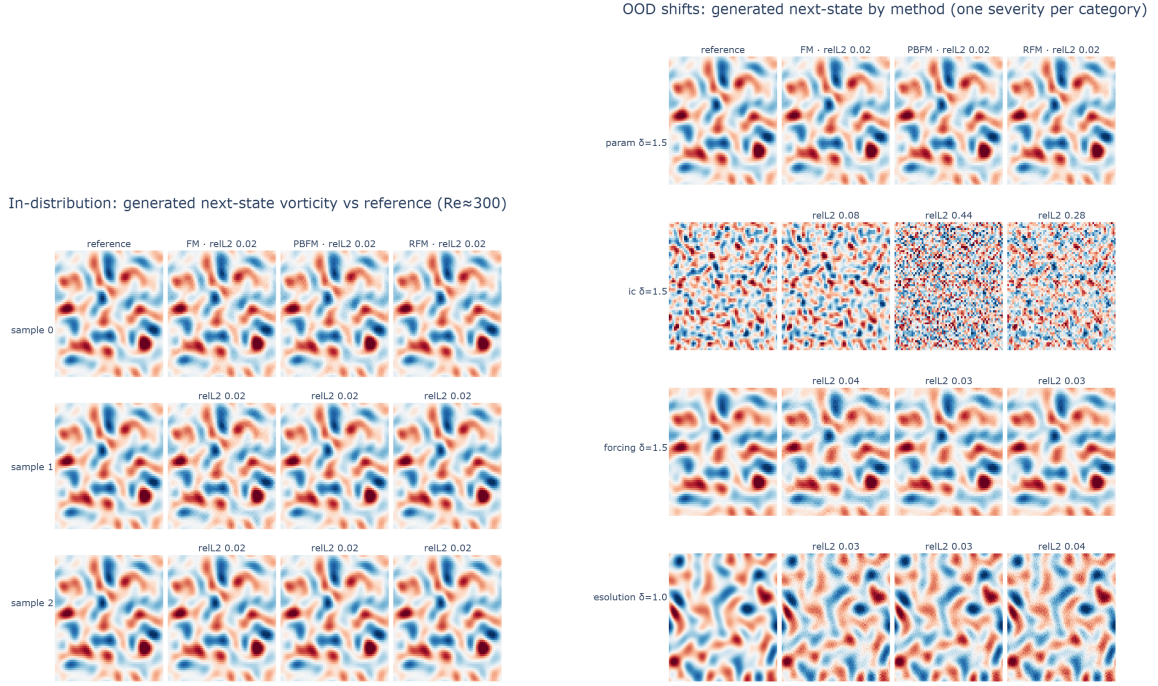


Figure 14: Generated next-state vorticity against the reference. Columns are reference, FM, PBFM and RFM, with the relative L2 to the reference printed on each generated panel. Observed: in distribution (left) the three models are close to the reference and to each other, with RFM sharpest at small scales. Under shift (right) the printed errors follow the table order, and the residual models place small-scale structure that the reference does not contain, which is the visual form of a model applying its training physics to inputs outside the training set.

Per-sample accuracy and physical consistency (lower is better)



Figure 15: Per-sample accuracy and physical consistency on identical inputs and noise. Observed: RFM has the lowest relative L2 (left) and the lowest PDE residual (right) on most of the four samples, and the dotted lines mark each method’s mean. The per-sample view agrees with the aggregate tables.

4.8 Summary of observations

1. In distribution the weighted residual improves accuracy, physical consistency and distributional fidelity together, by 37% to 63% depending on the axis, and reaches the baseline’s best accuracy with roughly 1/8 of the sampling budget.
2. The conflict-free method is not separable from the plain baseline on any in-distribution axis at 5 seeds. Its extra steps did not pay off in this setup.
3. The residual advantage is specific to the training distribution. Under ic and forcing shift the ordering reverses, the residual models become less accurate and less stable in rollout than the baseline, and PBFM is the weakest of the three.
4. The residual has a narrow usable weight band and collapses uniformly outside it.
5. Within this grid the ic and forcing shifts are the ones that separate the models: the param axis does not raise one-step error out to 3x the training Reynolds number, and the resolution axis moves in integer grid multipliers. The one-step breaking-point threshold of 0.3 is not reached at any severity, so the rollout horizon is the signal that separates them.

5 Conclusion

Within the range this study covers, a plain weighted residual is the better choice for a next-step Navier–Stokes operator in distribution, by a clear margin and at a large saving in sampling cost, while the parameter-free conflict-free method did not improve on the baseline. The benefit of the residual is local rather than general: it does not survive a move to higher-wavenumber initial conditions or stronger forcing, where it can make both accuracy and rollout stability worse than the plain baseline. A practitioner adding a physics residual to this kind of operator should tune its weight inside a narrow band and check robustness under the specific shifts they care about, rather than assume the in-distribution gain transfers.

Limitations

The results come from one setup, and several features of that setup bound what can be concluded. They are set out here as limitations rather than general claims, and where a cause is suggested it is offered as a possibility consistent with the observations, not as an established mechanism.

The conflict-free method is the clearest case. PBFM did not separate from the plain baseline on any in-distribution axis and was the weakest of the three out of distribution. This is a statement about the present setup, not about the method in general, which was proposed and validated on different problems, model sizes and training budgets [3]. Several features of this study could account for the result, and each is a limitation of the setup rather than a property of PBFM. The physics residual here is normalised by a ground-truth scale so that one weight stays comparable across the Reynolds range, and that normalisation reduces the magnitude of the physics gradient; because the conflict-free update balances the two gradients by direction and magnitude, a smaller physics gradient may leave the update close to the Flow Matching gradient, which would place PBFM near the baseline. The model is small, training runs for only 100 epochs, and the comparison uses 5 seeds, so a baseline that is already easy to match at this scale may hide a benefit that appears with a larger model or a longer schedule. The study also uses a single equation and a single residual form. The comparison is not perfectly controlled either: the PBFM branch runs in float32 without gradient clipping, while FM and RFM use mixed precision with clipping, so the null result folds in these two incidental differences as well.

The honest reading is that PBFM did not separate from the baseline in this small-budget, normalised-residual setting, not that the method fails.

The in-distribution advantage of the weighted residual is limited in a way the observations expose but do not fully explain. RFM is the most accurate in distribution yet the least accurate under the ic and forcing shifts and the least stable in the 128-step rollout, and its generated field carries extra high-wavenumber energy that the reference does not (Figure 6, Figure 13). One possibility consistent with these observations is that the residual sharpens the small scales in a way that helps in-distribution accuracy without matching the true small-scale dynamics, so the same sharpening could hurt once the input distribution moves and could accumulate over a long rollout. The measurements support this association but do not establish the mechanism, and a different residual form or weighting might behave differently. On this operator and this residual, the residual is better read as a small in-distribution accuracy gain than as a way to improve robustness.

The evaluation limits some conclusions as well. The param and resolution shifts moved the metrics little over the range tested, the one-step breaking-point threshold of 0.3 was not reached at any severity so that measure did not separate the models, and the seed spread for the residual models under ic is wide, so the size of the out-of-distribution reversal is measured with limited precision. Taken together, the findings should be read as specific to this operator, this residual and this budget, and may change with a larger model, a longer schedule, more seeds, or a different equation.

Next update

The planned continuation applies the same protocol to other PDEs, for example a diffusion-dominated and an advection-dominated problem, to measure whether the residual effect seen here is a property of 2D Navier–Stokes or changes from one equation to another. Running the same three models and the same shift grid on a second and a third equation would show whether the in-distribution gain, the out-of-distribution reversal and the narrow weight band hold across PDE families or are specific to this one.

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A The 2D Navier–Stokes system and the physics residual

The data and the residual use the vorticity form of the incompressible 2D Navier–Stokes equation on a periodic square of side 2π . With velocity $u = (u_1, u_2)$, scalar vorticity $\omega = \partial_x u_2 - \partial_y u_1$, kinematic viscosity ν and forcing f ,

$$\frac{\partial \omega}{\partial t} + (u \cdot \nabla) \omega = \nu \nabla^2 \omega + f, \quad \nabla \cdot u = 0, \quad \nu = \frac{1}{\text{Re}}. \quad (2)$$

The flow is incompressible, so velocity is recovered from vorticity through a streamfunction ψ with $\nabla^2 \psi = -\omega$ and $u = (\partial_y \psi, -\partial_x \psi)$. The forcing is a fixed Kolmogorov pattern with wavenumber $k_f = 4$ and amplitude $A = 1$ in the training regime. The solver is pseudospectral, with derivatives evaluated in Fourier space, and Re is mapped to viscosity through $\nu = 1/\text{Re}$.

The next-step residual measures how far a predicted next state is from satisfying this equation over one step Δt . Given the conditioning vorticity ω_t from u_t and the vorticity $\hat{\omega}$ of the predicted endpoint $\hat{x}_1$, the discrete transport residual is

$$R = \frac{\hat{\omega} - \omega_t}{\Delta t} + (\bar{u} \cdot \nabla) \bar{\omega} - \nu \nabla^2 \bar{\omega} - f, \quad D = \nabla \cdot \hat{u}, \quad (3)$$

where $\bar{\omega}$ and $\bar{u}$ are the time-scheme combination of the current and predicted states (the mid-point average in the runs here), and D is the incompressibility residual, which is zero for an exact divergence-free field. Both are computed in float32 for numerical stability. The physics loss normalises each term by a fixed ground-truth scale so that a single weight stays comparable across Reynolds number, and weights it linearly in the path time τ :

$$\mathcal{L}_{\text{phys}} = \mathbb{E} \left[\tau \left(\frac{R}{s_R} \right)^2 \right] + \alpha \mathbb{E} \left[\tau \left(\frac{D}{s_D} \right)^2 \right], \quad (4)$$

with s_R the RMS of the ground-truth tendency $(\omega_{t+\Delta t} - \omega_t)/\Delta t$, s_D the RMS of the ground-truth vorticity, and $\alpha = 1$. The residual is evaluated on the predicted endpoint $\hat{x}_1 = x_\tau + (1 - \tau) v_\theta$, not on an interior point of the path, because the interpolation between noise and data has no reason to satisfy the equation.

B Out-of-distribution mechanisms and severity

Each out-of-distribution family is a single-parameter shift of the data-generating regime, controlled by a severity $\delta \geq 0$. At $\delta = 0$ the regime sits at the edge of the training range, and larger δ moves it further from training. The four families change independent parts of the regime and are kept separate rather than combined into one score, because a model can be robust to one and fragile to another. Table 5 gives the exact maps, where $\text{Re}_{\text{max}} = 500$ is the largest training Reynolds number, $k_{\text{ic}} = 4$ the base initial-condition wavenumber, $A = 1$ the base forcing amplitude and $N = 64$ the base grid.

Two properties of these maps matter for reading the results. The param map raises Reynolds number at fixed forcing and grid, and the model is given the new Reynolds number through

Table 5: The four out-of-distribution shift families as a function of severity δ .

family	shifted quantity	map
param	Reynolds number	$\text{Re} = \text{Re}_{\max}(1 + \delta)$, $\nu = 1/\text{Re}$; $\delta=0 \rightarrow 500$, $\delta=2 \rightarrow 1500$
ic	initial-condition wavenumber	$k = k_{\text{ic}}(1 + \delta)$; a higher-frequency start state
forcing	forcing amplitude	$A' = A(1 + \delta)$; same pattern, stronger drive
resolution	grid size	$N' = (1 + \text{round}(\delta)) N$, $\Delta x = 2\pi/N'$; a quantised integer step

its conditioning, so within the range tested the fields stay qualitatively similar and the shift is mild. The resolution map multiplies the grid by an integer, so $\delta = 1$ and $\delta = 1.5$ round to the same grid and the axis moves in jumps rather than smoothly. Every out-of-distribution regime is normalised with the training statistics rather than its own, so that a genuine shift is measured rather than hidden by renormalisation.

C Model architecture

The velocity field is a conditional UNet operating on the two velocity channels of x_τ concatenated with the two channels of the conditioning field u_t , so four input channels. Diffusion time τ and Reynolds number are embedded and injected as feature-wise scale and shift at every residual block. Spatial derivatives used by the physics residual are computed in float32 for numerical stability regardless of the mixed-precision training of the network. Table 6 lists the sizes.

Table 6: Velocity network.

component	setting
backbone	conditional UNet, residual blocks with GroupNorm and SiLU
input channels	4 (2 state, 2 conditioning)
output channels	2 (velocity field)
base width	48
conditioning embedding	128 (diffusion time and Reynolds number)
conditioning mechanism	feature-wise linear modulation (scale and shift per block)
resolution	64×64 , periodic
parameter precision	float32 for the physics operators; network precision per model (Table 7)
weight averaging	exponential moving average with short warmup

D Reproducibility and hyperparameters

Table 7 lists the configuration used for every run in this study. Data are regenerated from the solver and seeds rather than stored, training resumes automatically from the last checkpoint, and every reported number is a mean over seeds with a bootstrap interval. Sampling uses Euler in all comparisons so that step count equals NFE.

Table 7: Configuration.

group	value
grid	64×64 , periodic torus, Kolmogorov forcing (wavenumber 4, amplitude 1)
time step	$\Delta t = 0.01$; warmup 50 steps; trajectory length 40
Reynolds (train)	$\{100, 200, 300, 400, 500\}$; evaluation at $\text{Re} = 300$
trajectories	6 per training regime, 2 per validation regime
model	UNet width 48, conditioning embedding 128
optimiser	AdamW, lr 2×10^{-4} , weight decay 10^{-4}
schedule	cosine decay over 100 epochs; batch size 32
precision, clipping	FM/RFM: mixed precision, gradient norm clipped to 1.0. PBFM: float32, no clipping
residual	full transport residual plus incompressibility on the predicted endpoint, normalised by ground-truth scale, linear time weighting
RFM weight	$\lambda = 0.005$ for the model comparison; sweep over $\{0, \dots, 0.1\}$
PBFM	closed-form two-gradient CONFIG update, 3 unrolling steps, time-weighted residual
sampling	Euler; budgets $\{1, 2, 4, 8, 16, 32, 64\}$; evaluation budget 16; 64 samples
model selection	best checkpoint by FM validation loss on the EMA weights (same for all three)
rollout	horizon 20, 8 trajectories, failure threshold 0.3
OOD	param, ic, forcing, resolution; severities $\{0, 0.5, 1, 1.5, 2\}$; train statistics reused
seeds	5 for the model comparison, 3 for the residual-weight sweep